Emergence of Norms and Co-operative Behaviour in a Partially Observable Multi-Agent Stochastic Game

Research Project Proposal

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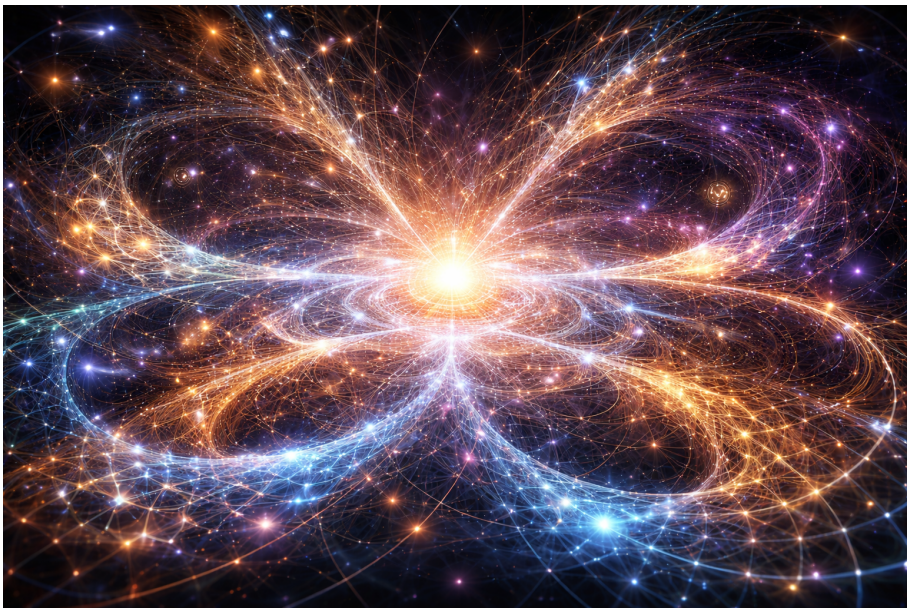

Abstract

We propose the *Initial Settler Problem*, a partially observable stochastic multi-agent environment designed to study the emergence, stability, and failure of simulated social norms under resource scarcity and epistemic uncertainty. A population of autonomous agents (number to be determined) inhabits a shared ecological system governed by a regenerating river whose long-term survival depends on the collective balance between harvesting from the river and investing into the river to regenerate it. Each agent receives only local, noisy observations from the environment (from its local patch of the river) and exchanges strategic, potentially deceptive messages regarding resource levels and the behaviour of other agents.

The environment is defined as a partially observable stochastic game with continuous ecological dynamics modelled via a logistic regeneration process and discrete, coupled action spaces that include actions affecting the environment and communication messages. Agents maintain internal belief states through recurrent policy architectures (we propose GRU), enabling them to compress historical observations, reputational signals, and simulated social interactions into a learned memory representation. Policy learning would be performed using independent proximal policy optimisation to address the non-stationarity induced by the continual change of the transition dynamics in the environment.

Across episodes (complete set of terminal states and episode length still needs to be determined), environmental conditions are reset while learned policies persist, allowing the investigation of "cross-generational" transmission (i.e. culture formation/habits) of behavioural patterns analogous to traditions, myths, and cultural norms in human society. We evaluate whether cooperation, truthful communication, and decentralised enforcement mechanisms emerge endogenously, and identify conditions under which these structures stabilise or collapse into exploitation, retaliation, or ecological failure (or perhaps other outcomes shall emerge).

This framework provides a testbed for studying how artificial agents form beliefs, assign trust, and exercise judgment in socially embedded, resource-constrained environments, with implications for long-term alignment, co-operative artificial intelligence, and the design of adaptive multi-agent systems interacting with human-focused environments.

1 Introduction

As artificial intelligence systems increasingly operate in real-world social and economic settings, their long-term safe impact depends not only on task performance, but on their ability to develop stable, interpretable, and socially compatible patterns of behaviour around humans.

The development of both embodied and digital artificial intelligence raises the problem of how generalisable intelligence can be acquired through interaction rather than through static, text-based or rule-driven supervision alone. Learning through direct engagement with dynamic, uncertain, and socially embedded environments may provide a foundation for agents to model human heterogeneity, adapt to diverse belief systems, and exercise context-sensitive judgment in human-focused settings.

From a brief search about emergence of norms in multi-agent systems, prior work has largely focused on behavioural convergence in static games (Morris-Martin et al. 2019, Cordova et al. 2024) or in preventing harmful emergent norms (Alqithami 2025) (detailed positioning relative to current research literature is still left to be done). It would seem that the interaction between norm emergence, reputation-based enforcement, and epistemic uncertainty in fully decentralised learning systems remains comparatively under-explored (a detailed positioning within the current research literature is left for future work).

In this paper, we would introduce a partially observable stochastic game framework that integrates a regenerating shared resource, fallible simulated social auditing, and persistent agent policies across episodes. This design enables the study of how co-operative behaviour, truthful communication, and decentralised enforcement mechanisms emerge, stabilise, or fail under long-term learning dynamics.

2 Problem Formulation

We formalise the *Initial Settler Problem* as a partially observable stochastic game (POSG) that models the interaction between autonomous agents and a shared, regenerating ecological resource (river) under epistemic uncertainty and decentralised, simulated social enforcement, wherein the agents themselves decide who to trust, accuse and punish. The environment is designed to capture both physical dynamics (resource regeneration, depletion, as well as distruction) and simulated social dynamics (communication, reputation, and fallible auditing), allowing the study of emergent norms, trust formation, and long-run behavioural stability or instability.

2.1 Game Structure

Let the environment be defined as a POSG, as follows:

$$\mathcal{G} = \langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}_{i\in\mathcal{N}}, \{\mathcal{O}_i\}_{i\in\mathcal{N}}, \tau, \{R_i\}_{i\in\mathcal{N}}, \gamma \rangle,$$

where,

- $\mathcal{N} = \{1, \ldots, n\}$ denotes the finite set of agents (settlers);
- $\mathcal{S}$ is the continuous, global state space;
- $\mathcal{A}_i$ is the action space of agent i;
- $\mathcal{O}_i$ is the observation space of agent i;
- $\tau(s_{t+1} \mid s_t, a_t)$ defines the transition probability process;
- R_i is the reward function for agent i;
- $\gamma \in [0, 1)$ is the discount factor.

We note that time evolves in discrete steps $t = 0, 1, 2, \ldots,$ and agents simultaneously act at each time step.

2.2 Hidden State

The global environment state at time t is defined as

$$s_t = (R_t, \{r_{t,i}\}_{i\in\mathcal{N}}, \{e_{t,i}\}_{i\in\mathcal{N}}, \{\rho_{t,i}\}_{i\in\mathcal{N}}),$$

where,

- $R_t \in [0, 1]$ is the global resource level of the river;
- $r_{t,i}$ is the local river segment associated with agent i at time t;

- $e_{t,i} \in [0, 1]$ is the private energy level of agent i at time t;
- $\rho_{t,i} \in \mathbb{R}$ is the public reputation score of agent i at time t.

We note that the global resource level is never directly observed by agents. Instead, each agent observes a noisy local approximation of its river segment.

2.3 Ecological Transition Dynamics

The river is modelled as a living system using a logistic regeneration process with agent-induced destruction and addition to the river's resource level:

$$R_{t+1} = \text{clip}\left(R_t + \alpha R_t(1 - R_t) - D_t + I_t + \varepsilon_t,\, 0,\, 1\right),$$

where,

- $\alpha > 0$ is the intrinsic regeneration rate (this is a hyper parameter which we will have to tune);
- $D_t = \sum_{i \in \mathcal{N}} D_{t,i}$ is the total damage caused by harvesting actions of the agents;
- $I_t = \sum_{i \in \mathcal{N}} I_{t,i}$ is the total investment contributed by agents;
- ε_t is the environmental noise.

We shall generate local river observations for each agent i at time t as follows:

$$r_{t,i} = R_t + \eta_{t,i}, \quad \eta_{t,i} \sim \mathcal{N}(0, \sigma^2),$$

where $\eta_{t,i}$ is a random variable used to introduce local environmental stochasticity into the agent's observation of the resource level.

An important note is that if our river's resource level, R_t, remains below a collapse threshold K for k (hyper parameters which shall be determined) consecutive time steps, the episode terminates, representing ecological failure.

2.4 Observations and Belief State

Each agent i shall receive an observation vector defined as follows:

$$o_{t,i} = \left(\tilde{r}_{t,i}, e_{t,i}, \{\rho_{t,j}\}_{j \in \mathcal{N}}, m_{t,i}, \mu_{t,i}\right),$$

where,

- $\tilde{r}_{t,i} = r_{t,i} + \xi_{t,i}$ is a noisy local river estimate (we add another source of noise here to account for the sensory error of the agent);
- $e_{t,i}$ is defined as above;
- $p_{t,j}$ is the reputation score of agent j that agent i has on file, including itself;
- $m_{t,i}$ is a bounded inbox of recent messages from other agents;
- $\mu_{t,i}$ is a private memory feature vector encoding trust estimates, past interaction information with other agents, and historical consistency of other agents' claims.

2.5 Action Space

Each agent selects a combined action, defined as:

$$a_{t,i} = \left(a_{t,i}^{\text{env}}, a_{t,i}^{\text{com}}\right),$$

where the action set affecting the environment is defined as:

$$a_{t,i}^{\text{env}} \in \{\text{HARVEST}, \text{INVEST}, \text{PUNISH}(j), \text{NOOP}\},$$

and the communication action set is defined as:

$$a_{t,i}^{\text{com}} = (\text{claim_level}, \text{accuse_target}, \text{charge}, \text{recommend}).$$

Messages allow agents to broadcast claims about the resource state and accusations regarding the behaviour of others, enabling strategic communication and deception.

2.6 Action-Environment Update Rules

The environment evolves as a direct consequence of agent actions. All state updates are clipped to preserve the physical constraints present in the envrionment: energy levels and resource levels remain within the bounded interval $[0, 1]$.

Harvest

When agent i harvests h units from the river (e.g. collecting water or fish), this action increases the agent's private energy while simultaneously damaging the ecological system. Energy conversion is governed by a harvesting efficiency parameter $\beta_h > 0$, such that:

$$e_{t+1,i} = \text{clip}\big(e_{t,i} + \beta_h\, h,\ 0,\ 1\big).$$

The corresponding environmental damage contribution shall be defined as:

$$D_{t,i} = \gamma_h\, h,$$

where $\gamma_h > 0$ controls the ecological cost of harvesting resources. In the baseline configuration, we initialise $\beta_h = \gamma_h = 0.1$, creating a symmetry between private gain and public loss. These hyper parameters should still be experimented with in order to explore how the system's dynamics evolve with adjustments of said hyper parameters.

Invest

When agent i invests v units of effort into maintaining or repairing the river (e.g. protecting spawning grounds or improving water flow), the agent incurs an immediate energy cost while contributing positively to the global resource level of the river.

The private energy update shall be given by:

$$e_{t+1,i} = \text{clip}\big(e_{t,i} - \beta_v\, v,\ 0,\ 1\big),$$

where $\beta_v > 0$ denotes the energetic inefficiency of investment. We intentionally set $\beta_v > \beta_h$ in the baseline configuration to make pro-investment behaviour locally costly and therefore strategically non-trivial. Once again, these hyper-parameters should be tuned to explore different long-term behaviour.

The corresponding environmental contribution to add back to the river's health is:

$$I_{t,i} = \gamma_v \, v,$$

with $\gamma_v > \beta_v$ in order to ensure that collective investment can, in principle, dominate collective extraction. In the initial regime, we set $\beta_v = 0.2$ and $\gamma_v = 0.3$, which could create behaviour analogous to structural tension between individual self-interest and collective benefit. (Same argument as above for tuning hyper parameters).

NoOp

The NoOp action represents rest or passive observation. This action would have no direct effect on the river and allows the agent to recover a small amount of energy:

$$e_{t+1,i} = \mathrm{clip}\big(e_{t,i} + g,\, 0,\, 1\big),$$

where $g > 0$ is a recovery constant.

An interesting experiment could be to test when $g < 0$, introducing a quasi-survival pressure in which inaction itself becomes costly. This transforms the environment into one where agents must continuously engage with the system, rather than simply abstain from participation. We are trying to prevent free-loading in this scenario.

Punish

When agent i selects Punish(j), a public audit mechanism is triggered, producing a noisy signal about whether agent j has misbehaved according to the relevant charge (e.g. greed, lying, or false accusation). This action imposes an energetic cost on both the enforcer and the target, reflecting the idea that social enforcement is itself costly, to prevent an agent from intentionally griefing another agent for its own benefit.

The energy update for the punishing agent shall be:

$$e_{t+1,i} = \mathrm{clip}\big(e_{t,i} - c_{\mathrm{pun}},\, 0,\, 1\big),$$

and for the punished agent:

$$e_{t+1,j} = \mathrm{clip}\big(e_{t,j} - c_{\mathrm{rec}},\, 0,\, 1\big).$$

In the baseline configuration, we initialise $c_{\mathrm{pun}} = 0.08$ and $c_{\mathrm{rec}} = 0.16$, ensuring that punishment is asymmetric and socially expensive (these hyper parameters need to be tuned as well...these are just estimate baselines). This asymmetry discourages indiscriminate or strategic griefing, while still allowing enforcement to emerge as a viable collective mechanism under appropriate conditions.

2.7 Social Auditing and Noise

To model epistemic uncertainty and the fallibility of social institutions (as is present within our reality), the environment would maintain a set of hidden behavioural metrics for each agent, corresponding to concepts such as greed, lying and false accusations. These metrics represent an underlying ground truth that is never directly observable by the agents themselves.

When agent i initiates a punishment against agent j, a public audit signal is generated and broadcast to all agents. This signal is defined as

$$y_t(i \to j) = M_{t,j} \oplus B_t, \quad B_t \sim \mathrm{Bernoulli}(p_{\mathrm{err}}),$$

where $M_{t,j} \in \{0, 1\}$ denotes the hidden ground-truth indicator of misconduct, and $\oplus$ is the XOR operator. This mechanism ensures that audits are neither perfectly reliable nor perfectly corrupt: with probability p_{err}, the public signal is flipped relative to the ground truth, introducing stochastic error present within the simulated social justice system.

This design should allow for three regimes to naturally arise: correct punishment, false punishment, and successful evasion (perhaps there are others that we shall see arise). We want to investigate if what emerges is a social system in which trust, enforcement, and reputation must be learned rather than assumed.

Greed (Over-harvesting and Free-loading)

A greed charge may be brought by agent i against agent j when j appears to extract disproportionately from the river relative to its contributions. We formalise this intuition through a harvest-investment ratio, which shall be defined as:

$$Q_{t,j} = \frac{h_{t,j} + \varepsilon}{v_{t,j} + \varepsilon},$$

where $h_{t,j}$ and $v_{t,j}$ denote the cumulative harvesting and investment actions of agent j over a fixed time window (perhaps this should be several previous time steps - still to be determined), and $\varepsilon > 0$ is a small constant to prevent division by zero.

We would consider agent j to be free-loading if:

$$Q_{t,j} > q_{\max},$$

where $q_{\max}$ is a hyper parameter. In the baseline configuration, we initialise $q_{\max} = 1.5$, reflecting the assumption that extraction is energetically easier than restoration.

The corresponding hidden ground-truth metric is defined as:

$$M_{t,j}^{\text{GREED}} = \mathbb{I}\big[Q_{t,j} > q_{\max}\big],$$

which remains hidden to the agents and is used only as an internal reference for our audit mechanism.

The public audit signal would be generated as:

$$y_t(i \to j) = M_{t,j}^{\text{GREED}} \oplus B_t, \quad B_t \sim \text{Bernoulli}(p_{\text{err}}).$$

This formulation allows guilty agents to occasionally evade detection and innocent agents to occasionally be falsely accused, reflecting the imperfect nature of our institutional judgment.

Lie (False Claims About the Environment)

To model deception in communication, the environment maintains a hidden metric indicating whether an agent's public claims about the river are consistent with subsequent environmental outcomes.

Let claims_j denote a set of resource-level claims made by agent j. After the next state transition, the environment checks whether the realised global resource level R_{t+1} corresponds to the claim made by agent j.

It follows that the ground-truth deception indicator shall be defined as:

$$M_{t,j}^{\text{LIE}} = \mathbb{I}\big[R_{t+1} \notin \text{claims}_j\big].$$

As before, agents do not observe this value directly. Instead, the public signal is generated as:

$$y_t(i \to j) = M_{t,j}^{\text{LIE}} \oplus B_t, \quad B_t \sim \text{Bernoulli}(p_{\text{err}}).$$

This construction ensures that honesty is not perfectly rewarded, nor dishonesty perfectly punished, forcing agents to reason under persistent uncertainty about the reliability of both others and the environment itself.

False Accusation

Finally, we allow the simulated social system itself to become an object of scrutiny. If agent i accuses agent j and the resulting audit clears j, the environment records this event as a potential false accusation by i.

Thus, the corresponding hidden metric is defined as:

$$M_{t,i}^{\text{FALSE}} = \mathbb{I}\big[\text{last accusation by } i \text{ failed audit}\big].$$

Once again, this noisy signal is passed through to the agents as:

$$y_t(i) = M_{t,i}^{\text{FALSE}} \oplus B_t, \quad B_t \sim \text{Bernoulli}(p_{\text{err}}).$$

This recursive structure allows the agents not only to reason about the behaviour of others, but also to reason about the integrity of the enforcement process itself. Over time, the system may develop stable norms around when to trust, when to accuse, and when to remain silent, or it may collapse into cycles of retaliation and institutional failure. These are all a priori hypotheses at this moment, and thus need to be tested.

2.8 Reward Function

Each agent should receive a per-step reward as follows:

$$u_{t,i} = w_f \Delta e_{t,i} - w_h \mathbb{I}[e_{t,i} \leq \lambda_h] - w_c \mathbb{I}[R_t \leq \lambda_c] - w_p \mathbb{I}[a_{t,i}^{\text{env}} = \text{PUNISH}],$$

where,

- $\Delta e_{t,i}$ is the change in energy due to the agent i's actions;
- λ_h is a survival threshold, which is a hyper parameter;
- λ_c is a collapse threshold for the river, which is a hyper parameter;
- w_f, w_h, w_c, w_p are adjustable weight hyper parameters.

This structure incorporates the selfish interest of survival and global ecological stability of the river, while allowing for the effect of social enforcement, which is costly and strategically non-trivial (perhaps there is a way to incorporate a negative reward if agent i has been punished...yet the agent may not necessarily have been guilty, so we would not want to reward incorrectly either).

2.9 Cross-Episode Learning and Study of Culture Formation

After each episode, we would reset the physical environment (river state and energy levels) while preserving learned policy parameters. Formally, let θ_i^l denote the policy parameters of agent i at the end of episode l. Then for the next episode, we have that:

$$\theta_i^{l+1} \leftarrow \theta_i^l,$$

while we re-initialise the state, s_0^{l+1}. This mechanism enables the study of behavioural patterns across episodes, which we could interpret as the emergence of cultural norms, traditions, and stable social strategies.

3 Learning Dynamics

Since the environment is partially observable, each agent must act on the basis of an internal belief about the world rather than direct access to the true global state. This belief must encode both the evolving condition of the river and the behavioural characteristics of other agents, as inferred from past interactions, communication, and social feedback.

Storing the full interaction history is neither scalable nor desirable as the number of agents, time horizon, and environmental complexity increase. Instead, each agent would need to maintain a compressed, adaptive memory that selectively retains information that proves useful for future decision-making and "forgets" what does not.

To implement this, from brief exploration online, we use a gated recurrent architecture (specifically, a GRU) as the agent's memory mechanism. The reason being is that we do not want the agent to remember the entire past, but rather, we want the agent to learn which aspects of past experience (such as trustworthiness of other agents, past interaction information with other agents, and historical consistency of other agents' claims) should be retained, and the rest forgotten.

Moreover, we are primarily interested in the long-run behavioural dynamics of the system rather than short-term performance. From the perspective of any single agent, the environment is inherently non-stationary, since the behaviour of every other agent is itself changing as learning progresses. Hence, the value of a specific action at one time step may differ from its value at a different time step.

Thus, we adopt a policy-based learning approach rather than a value-based one. Specifically, we consider utilising Proximal Policy Optimisation (PPO) to ensure that policy updates remain gradual and locally constrained. The reason we consider this algorithm is to avoid abrupt, unstable behavioural shifts and to instead encourage slow, gradual adaptation that more accurately resembles how agents could change their behaviour within a human environment.

This design would frame learning as a process of continued belief revision, rather than convergence to a fixed solution in a static world. We emphasize that here the goal is not necessarily to discover a single optimal policy, but rather to observe how patterns of behaviour, trust, and enforcement co-evolve under persistent uncertainty and shared resource constraints.

4 Research Questions

This project would investigate whether simulated social order, truth, and quasi-culture can emerge in a population of decentralised learning agents operating under partial observabil-

ity, ecological scarcity, and fallible institutions. Rather than forcing/training co-operative behaviour, we ask under which conditions it arises, stabilises, or fails.

1. Under which environmental, institutional, and incentive structures do co-operative norms (such as structures for harvesting, investment, and enforcement) emerge in a decentralised, partially observable multi-agent stochastic game?
2. How does noisy communication and fallible social auditing affect the long-run stability of punishment and reputation mechanisms? Is epistemic truth a viable equilibrium when deception yields a local advantage?
3. Can recurrent policy architectures (such as GRU) learn internal belief states that effectively mitigate the non-stationarity induced by simultaneously learning agents?
4. Which structural or behavioural failure settings lead to persistent dishonesty, retaliatory cycles, or irreversible societal collapse?
5. When the physical environment resets but agent policies persist, do stable behavioural patterns continue across episodes in a manner analogous to cultural formation? If so, is it possible to observe stable strategies (analogous to traditions), persistent false beliefs (analogous to myths), and the ability of agents to evolve their own moral framework (when to punish, when to be truthful, a sort-of 'what is right', 'what is wrong' according to the environment)?

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